KAPRKAR CONTEST – FINAL - SUB JUNIOR

AMTI-OCTOBER 26, 2002

Instructions:
1. Attempt as many questions as possible.
2. Elegant solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Attach the filled-in Face Slip and your rough working sheets along with the answer script.
5. Maximum time allowed is THREE hours.

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1. Rekha was asked to add 14 to a certain number and then divide the result by 4. Instead she first added 4 and then divided by 14. Her result was 5. Had she followed the instructions correctly, by how much would her result have differed from the incorrect result?

2. In the following display each letter represents a digit.

3	*B*	*C*	*D*	*E*	*8*	*G*	*H*	*I*

If the sum of any three successive digits is 18, find the value of ***H***.

3. Find an integer n that leaves remainders 2, 3, 4 when divided by 3, 4, and 5 respectively.

4. Find the number of two digit numbers whose sum of the digits is a single digit number.

5. Find the *sum* of *all* the digits of the result of the subtraction $10^{99} - 99$.

6. Rengu and Dingu are friends. One of them lies on Mondays, Tuesdays and Wednesdays and tells the truth on the other days of the week. The other fellow lies on Thursdays, Fridays and Saturdays and tells the truth on the other days of the week. On a particular noon, the two had the following conversation:

   Rengu : I lie on Saturdays
   Dingu : I will lie tomorrow
   Rengu : I lie on Sundays

   On which day of the week did this conversation take place?

7. About how many lines can one rotate a regular hexagon through some angle $x^\circ (0^\circ < x^\circ < 360^\circ)$, so that the hexagon returns to its original position?

8. In the given addition, each letter stands for a natural number. (Identical letters denote the *same* number). Find the number for *each* letter.

$$\begin{array}{r} ONE \\ +\underline{FOUR} \\ FIVE \end{array}$$

9. The sum of the lengths of the three sides of a right triangle is 18. The sum of the squares of the lengths of the three sides is 128. Find the area of the triangle.

10. In a trapezium ABCD, AB $\|$ CD and $\angle D = 2\angle B$. If DC $= p$ and AD $= q$ find AB.

11. A rectangle has sides of integer length (in cm) and an area of 36 cm^2. What is the maximum possible perimeter of the rectangle?

12. Appu has 6 marks (all integers) on his Report Card.
   i) The mean of the 6 marks is 74.
   ii) The mode of the 6 marks is 76.
   iii) The median of the 6 marks is 76.
   iv) The lowest mark is 50.
   v) The highest mark is 94.
   vi) Only one mark appears twice and no mark appears more than twice.

   Find the number of possibilities for his second lowest mark.

13. The natural numbers from 1 to 2100 are entered sequentially in 11 columns, with the first 3 rows as shown.

	Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8	Column 9	Column 10	Column 11
Row 1	1	2	3	4	5	6	7	8	9	10	11
Row 2	12	13	14	15	16	17	18	19	20	21	22
Row 3	23	24	25	26	27	28	29	30	31	32	33
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

If the number 2002 occurs in column p and row q, find the value of $p+q$.

14. Find the sum of all the coefficients of the polynomial $(x-2002)^3(x-2001)^2(x-2000)(x-3)^3(x-2)^2(x-1)$.

15. Find the natural number n such that $2^{13}+2^{10}+2^n$ is a perfect square of an integer.

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